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Professional Experience:

Data Science Intern | 10 Pearls | 11/2024 – 01/2025

- Developed an **end-to-end AQI prediction system**, improving forecast accuracy by **82%** using **Random Forest, Ridge Classification, and Logistic Regression**.
 - Engineered a **feature pipeline** integrating real-time weather and pollutant data from OpenWeather API, optimizing data processing efficiency.
 - Built an **interactive Streamlit web app** for real-time AQI visualization, enhancing user engagement.
 - Automated **data pipeline** using **GitHub Actions** and **Hopworks Feature Store**, ensuring seamless model retraining.
 - Integrated **SHAP/LIME** for explainable AI and **Seaborn** for data visualization.
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Projects:

Gradient Descent (Python)

Implemented **linear regression with gradient descent**, optimizing convergence speed. Developed **cost function visualizations** using Plotly.

Timetable Generation Using Graph Algorithms (Python, Java)

Developed an **automated scheduling system**, reducing timetable conflicts by **90%** using **graph-based maximum matching and graph coloring**. Exported **color-coded schedules** to Excel for improved usability.

Divide and Conquer Algorithm Implementation (React, Python)

Designed a system implementing **Closest Pair of Points** and **Integer Multiplication** using Divide and Conquer. Built an **interactive React UI** for dataset selection and result visualization.

Skills:

Programming & Data Science:

- **Languages:** Python (Intermediate), SQL (Intermediate), Java (Basic), C/C++ (Advanced)
- **Machine Learning:** Scikit-learn, PyTorch, XGBoost
- **Data Visualization:** Seaborn, Matplotlib, Plotly, SHAP, LIME

Big Data & Databases: MySQL, MongoDB

Cloud & Deployment: Google Cloud, Docker, GitHub Actions

Productivity & Tools: Git, Linux, Jupyter Notebook, Google Colab

Academic Qualification:

National University of Computer and Emerging Sciences | 2022 – Present

Bachelor of Science in Computer Science | CGPA: 3.76

- **Relevant Coursework:** Deep Learning, Artificial Intelligence, Probability and Statistics, Numerical Computing, Parallel Distributed Computing, Data Structures & Algorithms.
 - **Achievements:** Three-time **Dean's List** achiever and one-time **Rector's List** achiever
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